

Photocatalytic Air Purification

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Abstract: Titanium dioxide represents an effective photocatalyst for destruction of volatile organic compounds (VOC), NO_x and SO_2 in indoor and outdoor air. The mechanism of selected pollutants removal as well as the dependence of reaction rate on some key influencing factors are discussed. Recent papers and patents considering photocatalysts preparation and immobilization techniques including type of supports, surface pretreatment procedure are reviewed. Photocatalytic reactors are designed in order to provide strictly controlled conditions for photocatalytic air purification. Some reactors can be utilized in commercial applications as a part of HVAC systems (Heating, Ventilation and Air Conditioning) but most of them are used in laboratories to measure the activity of different types of photocatalysts applied for gas streams treatment. Examples of photoreactors used for gas phase treatment are presented.

Keywords: Air treatment, Air purification, Heterogeneous photocatalysis, Titanium dioxide, Gas phase infiltration.

1. INTRODUCTION

Air pollutants such as chemicals, particulate matter and biological materials cause harm and discomfort to humans and other living organisms. Major chemicals, known as air pollutants and produced by human activity, include: sulfur dioxide (SO_2), nitrogen oxides (NO_x), volatile organic compounds (VOCs). Exposure to these pollutants is associated with numerous effects on human health, including increased respiratory symptoms, hospitalization for heart or lung diseases, and even premature death. Nitrogen oxides are responsible for urban smog because of their photochemical reactions with hydrocarbons. Together with sulfur dioxide and sulfur trioxide they are major contributors to the acid rain. Volatile organic compounds refer to organic chemical compounds which have significant vapor pressures and which can affect the environment and human health. VOCs are numerous, varied, and ubiquitous. VOCs include both man-made and naturally occurring chemical compounds, such as solvents (aliphatic hydrocarbons, ethyl acetate, glycol ethers and acetone), chlorofluorocarbons, chlorocarbons, formaldehyde and MTBE (methyl tertiary butyl ether).

In recent years, various air-purifying systems using TiO_2 photocatalysts have been commercialized. Photocatalytic oxidation has gained much attention in the purification of air because it provides an attractive alternative to the more conventional approaches such as carbon adsorption, incineration or catalytic oxidation. The main advantages of heterogeneous photocatalysis over other oxidation treatment result from the direct absorption of light which allows degrading of the pollutants at ambient temperature, pressure, humidity, etc. [1]. Additionally, low concentration and low flow rate waste streams can be easily processed. Weak UV irradiation of less

than 1 mW/cm^2 , which is needed for the photoconversion, is excessively provided by solar light (up to 4 mW/cm^2). Moreover, no extra reactants such as NH_3 or O_3 are required [1].

The mechanism of TiO_2 excitation, as well as reactions occurring at TiO_2 surface, are well described in the literature [2-4]. While a photon with an energy of $h\nu$ matches or exceeds the band gap energy, E_g , of semiconductor, an electron (e^-) is promoted from the valence band (VB) into the conduction band (CB) leaving a hole (h^+). Excited electrons and holes can recombine, get trapped in metastable surface state, or the redox potential for photogenerated holes is $+2.53 \text{ V}$ versus the standard hydrogen electrode in pH 7 solution [3]. After reaction with water, these holes can produce hydroxyl radicals ($\cdot\text{OH}$), whose redox potential is only slightly decreased. The redox potential for CB electrons is -0.52 V , which is negative enough to reduce O_2 to superoxide, or to hydrogen peroxide. Depending on the reaction conditions, the holes, $\cdot\text{OH}$, O_2^- , H_2O_2 and O_2 itself can all play important roles in the photocatalytic reaction mechanism [3].

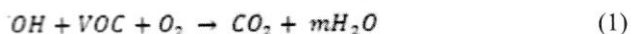
In this paper, the recent patents and research papers pertaining to photocatalyst and photoreactors used for gas phase purification including various types of photocatalyst preparation and immobilization methods are presented. The mechanism and efficiency of selected pollutants degradation in the presence of TiO_2 are also discussed.

2. VOCs PHOTODEGRADATION

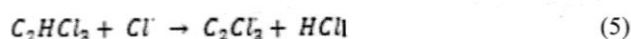
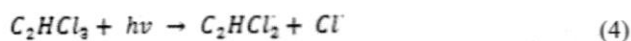
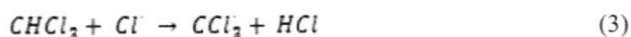
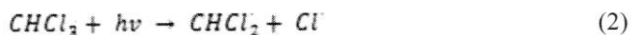
The efficiency of VOCs photooxidation depends on a few factors such as: temperature, humidity, light wavelength/ intensity, oxygen concentration, contaminant structure and concentration, surface velocity, resident time, photoreactor geometry, photocatalyst type and loading. The effect of the UV-intensity, initial concentration of gas phase pollutants and water vapor for VOCs removal was comprehensively described by Zhao & Yang [5] and Mo *et al.* [6].

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Volatile organic compounds could be chemically transformed by the hydroxyl radical created at the surface of irradiated photocatalyst. The summary reaction between $\cdot\text{OH}$ radical and VOCs can be expressed as:

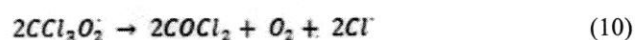


Kinetics and mechanism of UV-photodegradation of chlorinated organics in the gas phase was investigated by Feiyan *et al.* [7]. Photodegradations of three chlorinated VOCs: chloroform, carbon tetrachloride (CTC) and trichloroethylene (TCE) were investigated in a semi-batch reactor using a low-pressure mercury lamp. They proposed that $\text{Cl}^\cdot$ initiated/ sensitized oxidation is the reason for the increased degradation rate of TCE and the shift in the reaction order for chloroform. The amount of $\text{Cl}^\cdot$ generated by the direct photolysis of chlorinated compounds is the function of the initial VOCs concentration and the reaction is enhanced in the presence of reactive hydrogen atom, according to the following reactions:



In spite of undergoing $\text{Cl}^\cdot$ sensitized degradation, the differences in degradation shown by chloroform and TCE are due to their structural difference and consequently due to the difference in their UV absorption spectra. Chloroform does not show $\text{Cl}^\cdot$ sensitized degradation until above 500 ppm, whereas TCE shows the effect from the lower concentration (i.e. 80 ppm). Chloroform does not absorb at 254 nm and its molar extinction coefficient at 185 nm is much lower (692 L/mol·cm) than that of TCE ($\sim 10^4$ L/mol·cm) [7]. Since TCE absorbs only slightly at 254 nm (~ 10 L/mol·cm), the abundance of 254 nm photons makes TCE degradation rate quite high.

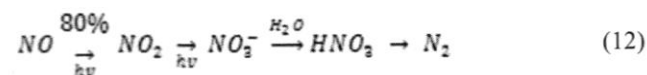
Chlorine radicals do not react with the C-Cl bond, hence CTC may not undergo a $\text{Cl}^\cdot$ sensitized reaction. The following mechanism was proposed for carbon tetrachloride degradation:



3. NO_x AND SO₂ REMOVAL

A number of studies have been published to utilize UV irradiation for direct photocatalytic degradation of NO into N₂, N₂O, NO₂, and O₂ over semiconductor and zeolite catalyst at ambient temperature [1, 8, 9]. It has been reported that the decomposition reaction of NO can be proceeded photocatalytically on powdered TiO₂ at room temperature. Additionally, TiO₂ photocatalysts have been practically applied for soundproof highway walls to eliminate the NO emitted from vehicle's engines. Table 1 shows the efficiency of NO_x removal at the surface of powdered and immobilized TiO₂.

When the TiO₂ photocatalyst is irradiated by solar light in the presence of NO in atmospheric conditions, NO is oxidized into NO₂ and subsequent oxidized into NO₃⁻. The NO₃⁻ species created at the TiO₂ surface can be removed as HNO₃ by water such as rainwater. Schematic reactions for the photocatalytic transformation of NO molecules are shown below:



Reaction between SO₂ and HO₂ radicals could cause SO₄²⁻ ion formation, as shown in Eqs. 14-17.



The effect of several process variables on NO and SO₂ removal in the non-thermal plasma-TiO₂ photocatalysts hybrid system was investigated by Nasonova *et al.* [10]. The TiO₂ thin film was coated on the glass beads uniformly without cracks by a rotating cylindrical PCVD reactor. It was observed that the increase of the initial NO and SO₂ concentrations or the decrease of the gas residence time resulted in the decreased removal efficiencies of both NO and SO₂. However, the NO and SO₂ removal efficiencies increase with the increase of the applied pulse frequency. It was found that the removal efficiency of NO and SO₂ were almost 100% for the following experimental conditions: [NO]₀ = 260 ppm, [SO₂]₀ = 250 ppm, f = 900 Hz, and τ_r = 1 s, where f is a pulse frequency and τ_r is a residence time of gas stream.

A TiO₂-containing paint was used in an indoor car park in order to test the de-polluting efficiency [11]. The ceiling surface of the car park (322 m²) was covered with white acrylic paint treated with 10% TiO₂ provided by Millenium Chemicals, see detail in Table 1. The close area (917 m³) was fed with car exhaust gases and irradiated by UV lamps for 5h. Illumination was provided by 20 UV lamps (MAZDA/36 W/2500 lm), which were fixed symmetrically, 40 cm from the ceiling. The photocatalytic removal of NO and NO₂ was calculated to 19% and 20%, respectively, while

Table 1. Efficiency of NO_x Photoconversion

Pollutant Type	Initial Concentration of Pollutants	Experimental Condition	Type and Form of Photocatalyst	Conversion Product	Conversion (%)	Ref.
NO	1365 µg·m ⁻³	close area (917 m ³) fed with car exhaust gases and irradiated by 20 UV lamps (MAZDA/36 W/2500 lm) for 5h	white acrylic paint treated with 10% TiO ₂ (Millenium Chemicals)	no data	19	[11]
NO ₂	1679 µg·m ⁻³			no data	20	
NO	200 ppm	Photoreactor volume = 18.6 dm ³ 6W UV lamp - Cole-Parmer)	TiO ₂ prepared by sol-gel supported on glass fiber filter (Whatman)	NO ₂	95.5	[12]

the photocatalytic rate ranged from 0.005 to 0.13 µg·m⁻²·s⁻¹ for NO and from 0.09 to 0.16 0.13 µg·m⁻²·s⁻¹ for NO₂. The same material was tested in an environmental chamber, under strictly controlled environmental conditions. NO (or NO₂) was the only pollutant in the chamber, while the temperature and relative humidity were set at 23°C and 20%, respectively. The photodegradation rate was 0.18 µg·m⁻²·s⁻¹ for NO, which indicates better photocatalytic behavior of the material under laboratory scale experiment. The author mentioned, that the controlled reaction conditions of a lab scale test result in less inhibition effects, which usually leads to an overestimation of the photocatalytic capacity of an TiO₂-containing material. The presence of organic pollutants in exhaust gases decreased the photocatalytic efficiency of the paint to oxidize NO_x.

4. PHOTOCATALYSTS AND PHOTOCATALYST IMMOBILIZATION (COATING)

4.1. Photocatalysts

Different photocatalysts have been used for outdoor and indoor air purification. Photocatalysts are semiconductors and commonly belong to the group of metal oxides or sulfides. For example, the photocatalytic materials can be TiO₂, ZnO, WO₃, SnO₂, ZrO₂, Fe₂O₃, V₂O₅, FeTiO₃, ZnS and CdS [6, 13, 14]. Whereas, the most widely investigated and used photocatalyst is TiO₂. The advantages of TiO₂ are that it is highly photocatalytic active, non-toxic and durable [15]. But also other photocatalysts like tungsten oxide find technical application [16]. Table 2 summarizes photocatalysts for air purification described in recent papers and patents. They are used to purify air from volatile organic compounds (aldehydes, ketones, alcohols, aromatics, alkenes, alkanes), odors, pathogens, mildew, NO, CO and O₃. The photocatalytic process and air purification can be improved by several methods. The photocatalyst can be doped or modified with metal ions, metal oxides, noble metals or non-metals. Metal oxides, noble metals and non metals like N, S, C are reported to make TiO₂ responsive for visible light [16-20]. Noble metals like Au, Pt and Pd reduce the electron hole recombination [21], amend the decomposition rate of contaminants [22, 23], act as thermocatalysts for CO oxidation [21] and prevent the deactivation of photocatalyst [23]. Furthermore, composite materials exist. TiO₂ is combined with absorbing materials like SiO₂ or Al₂O₃, which increase the surface area, adsorb contaminants and prevent the catalyst deactivation by

siloxanes [23, 24]. MnO in TiO₂/MnO composites decompose O₃ [21]. Additional metal oxides in Au-TiO₂ catalysts are considered to prevent the migration of Au nanoparticles. And WO₃/TiO₂ catalysts are less sensitive to humidity variations [25]. A comprehensive summary of various photocatalysts is given in the literature review by Mo *et al.* [6]. They summarize several synthesis methods, like gas phase and liquid phase methods, and the photocatalytic improving. Yu *et al.* in their research review, summarize the existing methods of improving the degradation efficiency and utilization of visible light [26].

4.2. Supports for Photocatalysts

To attach the photocatalyst onto a support is essential for running a photocatalytic air purifier. Since the activity of photocatalyst is directly proportional to the surface area, nanomaterials are used. Without immobilization, the photocatalyst would be simply blown away and lost or the photocatalyst has to be recovered downstream by a dust collector. Dust collectors are economically inapplicable, since with decreasing particle size the deposition effort increases. Immobilization and coating of photocatalysts on appropriate supports is a scantily discussed topic in current literature reviews [5, 6, 26]. Table 3 summarizes supports and their shape used in laboratory and for practical/patented applications. The ideal support should have the following characteristics: a) strong bonding of the photocatalyst to the surface without negatively affecting their reactivity, b) high specific area, c) support is not decomposed under UV irradiation, d) good adsorption capability for compounds to be degraded, e) facilitate mass transfer processes, and f) chemical inert. Furthermore, it should be economically and technically realizable. Organic materials should not be used, since they are altered photochemically [30]. If glass is used, attention has to be paid to the UV transmission rate. Borosilicate glasses (e.g. Pyrex glass) have good UV-A and UV-B transmission [31]. Coating of asphalt, concrete and plaster were especially developed for outdoor air treatment [32-34]. Ramirez *et al.* investigated the weathering resistance of coated plasters and concretes [33]. The three weathering stages are not specified in detail, but they find that porous and rough materials retain more TiO₂ particles after dip-coating. The highest TiO₂ deposition rate was obtained for plasters since their pores and cracks have bigger size compared to the concrete samples. The deposition was 37% - 55% for plasters and 17% for white cement with limestone, respectively. Aluminum or

Table 2. Composition of Photocatalysts and Their Application in Gas Phase Purification Technologies

Photocatalyst	Promoting Effect of Additional Component	Irradiation	Contaminants	Source ¹	Reference
P25	-	365 nm	NO, benzene, toluene, ethylbenzene, o-xylene	A	[27]
P25	-	400 – 700 nm, main peak at 365 nm	Microorganisms (<i>E. coli</i> , <i>Pseudomonas fluorescens</i> , <i>Microbacterium sp.</i> , <i>B. Subtilis</i> , <i>Microbacteriaceae str.</i> and <i>Paenibacillus sp.</i>)	A	[28]
TiO ₂ /Al ₂ O ₃	Adsorbing material	(290 nm ≤ λ ≤ 420 nm)	NO	A	[24]
TiO ₂ /InVO ₄	Visible light responsive	(450 nm ≤ λ ≤ 900 nm)	benzene	A	[18]
C–N–S–TiO ₂	Visible light responsive	Solar light	NO	A	[19]
Pt–TiO ₂ /SiO ₂	Adsorbing material and increased decomposition rate	300 – 400 nm	acetaldehyde	A	[22]
WO _x /TiO ₂ /Fe _x O _y	Increased quantum yield	Visible light	Dirt, bacteria, mildew	P	[16]
WO ₃ /TiO ₂	Low sensitivity to humidity variations	180 – 400 nm	VOCs and semi-VOCs	P	[25]
Multi layer: Me/TiO ₂ /MnO/TiO ₂ (Me=Au or Pt)	Metal nanoparticles prevent recombination and act as thermo-catalyst for CO oxidation, MnO prevents migration of Au and decomposes O ₃	180 – 400 nm	CO, O ₃ , low polarity organic compounds, VOCs, semi-VOCs	P	[21]
One or more layers: TiO ₂ , metal clusters, doped semiconductor, noble metals	Improved decomposition, prevents deactivation	UV	VOCs, siloxanes	P	[23]
Multi layer: TiO ₂ /Me/Au/MnO _x (Me = Rh, Ru, Pd, Ir, Os, Pt, Cu, Ag, Re)	Metal nanoparticles prevent recombination, MnO _x decomposes CO and O ₃	UV	VOCs, CO, O ₃	P	[29]

¹A: article; P: patent

Table 3. Supports for Photocatalysts

Support	Shape	Photoreactor Type	Application	Scale of Application	Reference
Glass fiber	Fiber	Fixed bed	Indoor	Laboratory	[12, 27, 37-39]
Stainless steel	Flat plate	Fixed bed	Indoor/outdoor	Laboratory	[40, 41]
Silica gel	Spheres (ø 2.8 - 3.35 mm)	Fixed bed	Indoor	Laboratory	[22]
Pyrex glass	Tubular	Fixed bed (annular flow)	Indoor	Laboratory	[42-44]
Nickel wafer	Foam	Fixed bed	Indoor	Laboratory	[45, 46]
Asphalt	Road	Fixed bed	Outdoor	Laboratory	[34]
Concrete and plaster	Flat	Fixed bed	Outdoor	Laboratory	[32, 33]
Ceramics	Honeycomb monolith	Fixed bed	Indoor	Laboratory	[47]
Aluminium	Honeycomb	Fixed bed	Indoor	Laboratory	[30]

(Table 3) Contd....

Support	Shape	Photoreactor Type	Application	Scale of Application	Reference
Aluminium or Aluminium alloys	Array of voids	Fixed bed	Indoor	Patented system	[21, 23, 25, 29, 35]
Aluminium, copper, steel, stainless steel, FeCrAlY alloy, ceramics	woven, honeycomb, fins, mesh, a fibrous structure, or a filamentous structure	Fixed bed	Indoor	Patented system	[36, 48]
Window pane, plastic, concrete, wall material, tents, unwoven cloth	-	Fixed bed	Indoor/Outdoor	Patented system	[49]
Fiberglass, synthetic or natural organic fibers	Fibrous non woven or honeycomb	Fixed bed	Indoor	Patented system	[50]
Metal oxides	Particles	Fluidized bed	Indoor	Patented system	[51]
Not specified	Tubular	Fixed bed (annular flow)	Indoor	Patented system	[52]

aluminum alloys are often mentioned as support in patents [21, 23, 25, 29, 35, 36]. The advantages are that it has a low density, a high strength to weight ratio and is ductile, that is why it can be easily drawn, cast, machined and extruded. In the patent of Garfield and Potter [36], aluminium is preferably used, since TiO_2 adheres stronger to aluminium than other comparable materials with strength characteristics like aluminium. Taranto *et al.* exposed their aluminium honeycombs to mechanical strains before photocatalytic usage and determined losses lower than 6 wt% [30]. And during the photocatalytic air treatment (volume flow of $50 \text{ m}^3 \text{ h}^{-1}$ or air velocity of 0.34 m s^{-1}), the TiO_2 losses were estimated in the order of 1 wt% [30]. But also other support materials are possible like ceramics, metal oxides, metals and metal alloys. The predominant support shapes are array of voids and fibrous structures (see Table 3). Whereas array of voids prevail in patents. These arrays of voids have honeycomb structure, where the hexagonal design seems to be most cost effective, but also square, rectangular or triangular designs are possible [23].

4.3. Surface Pretreatment and Binders

The adherence of photocatalyst is a crucial prerequisite and can be improved by prior conditioning of the support surface and by the usage of binders. The methods are summarized in Table 4. Oils, greases and contaminants are removed from metal and glass supports by washing with detergent waters, alcohols or acetone. The washing can be improved by the usage of ultrasonic wave baths. Matsui and Tagata [49] have patented a special wiping method, where the glass surface is cleaned by a dirt-removal-agent which impregnates the wiping material. Another method for removing organic contaminants was reported by Sarantopoulos *et al.* [53]. They heat-treated their glass substrates at 500°C in air to oxidize organic contaminants. For metal, concrete or ceramic materials, grinding and blasting are the methods for removing dirt and corrosion layers. Passivation layers on metals can be removed by etching the surfaces [30,36]. Garfield and Potter [36] in their patent describe a special aluminium surface treatment. At first, the surface is washed

with detergent solution and etched to remove metal oxides. Subsequently, the surface is rinsed with water and immersed in a chromate or phosphate solution to form a conversion coating. Finally, it is rinsed in demineralized water to reduce the presence of salts. These salts have been found to cause corrosion. The conversion coating prevents the formation of aluminium oxides and reduces the incidence of film corrosion. The aluminium honeycomb walls have to be thick enough to withstand the etching process. Taranto *et al.* observed mechanical weakening for $64 \mu\text{m}$ thick honeycomb walls even for diluted acids (HCl , HNO_3 , H_2SO_4 or H_3PO_4).

Additional compounds can be spread onto the support surface or mixed with the photocatalyst nano-powder to increase the adherence of photocatalyst. Such compounds are: silanes, resins, carbowaxes and SiO_2 . SiO_2 nanoparticles are deposited on fibrous materials binding better TiO_2 particles to the fibers [30, 50]. If the photocatalyst is applied by chemical vapor deposition, SiO_2 layers have been found to prevent diffusion of impurity atoms and to influence the deposition [40]. Chen and Liu [34] modified TiO_2 with silane coupling reagent and sprayed it on asphalt surfaces. Kim and Choi [54] mixed the polyethylene glycol carbowax with TiO_2 powder in aqueous solution, casted it on the glass surface and heat-treated in air at 450°C for 30 min to burn out the organics. The utilization of resins is described in patents. In Garfield and Kirk [36] method TiO_2 is mixed with 30 wt% polymer resin. The TiO_2 /resin mixture is electrostatically charged and sprayed onto the surface. Subsequently, the substrate is cured in an oven. The charge prevents moving of the particles during the heat-treatment. The cured photocatalyst is resistant to chipping and flaking for many years. Matsui and Tagata [49] apply the photocatalyst and resin by two different methods dependent on the substrate. For tile and non-transparent outside wall materials, the resin binder is firstly spread on the surface and then the photocatalyst is sprayed on the half-dried surface. The performance of the catalyst is not changed when the catalyst is baked on the substrate at temperatures of 400°C or less. For glass surfaces, the resin binder and the photocatalyst mixture are wiped with unwoven cloth onto the support.

Table 4. Surface Pretreatment and Usage of Binders for Various Supports Reported in Literature and Patents

Support	Immobilization	Binder	Surface Pre-treatment	Source*	Reference
Aluminium	Dip-coating	-	1) Ultrasonic wave bath in distilled water 2) Diluted acid (HCl, HNO ₃ , H ₂ SO ₄ or H ₃ PO ₄)	A	[30]
Stainless steel	CVD**	SiO ₂	1) Washing with warm water and detergent 2) Rinsing with water 3) Rinsing with isopropanol 4) Rinsing with distilled water	A	[40]
Steel and stainless steel	Plasma spraying	-	Sandblasting	A	[41]
Foam nickel wafer	Sol-gel	-	1) Ultrasonic wave bath in ethanol 2) Washing with distilled water and drying at room temperature 3) Calcination at 500 °C for 10 min in air	A	[45, 46]
Concrete products	Suspension coating	-	Sandblasting or grinding	A	[32]
Asphalt	Atmospheric plasma spraying	Silane KH-570 (γ-Methacryl-oxypropyl-trimethoxysilane)	-	A	[34]
non-woven cellulose fibers	Dip-coating	SiO ₂	-	A	[30]
Flat borosilicate glass, "E-glass" type tissue	CVD**	-	Thermal treatment at 500 °C in air	A	[53]
Pyrex glass tube	Sol-gel	-	Ultrasonic wave bath in isopropyl alcohol	A	[44]
Glass plate	Suspension coating (cast)	50% carbowax (polyethylene glycol) in aqueous solution	-	A	[54]
Aluminium or aluminium alloys	Suspension coating	-	Cleaning with acetone and methanol	P	[25, 29]
Aluminium	Electrostatic spraying	Polymeric resins (terephthalate ester-based resin: 2,2-dimethylpropanediol, polyethylene terephthalate)	1) Cleaning with detergent solution 2) Etching 3) Dipping in chromate or phosphate solution 4) Rinsing with demin. Water	P	[36]
Metal, ceramics	Shot peening	-	Blasting	P	[14]
Glass	Wiping material soaked with photocatalyst	Silicon resin, acrylic fiber resin, fluorine resin, epoxy resin	Wiping material with cleaner (ethanol, surface-active agent, tocopherol, ion exchange water, L ascorbic acid, citric acid)	P	[49]
Concrete block, outside wall material	Spraying	Silicon resin, acrylic fiber resin, fluorine resin, epoxy resin	-	P	[49]
Fiberglass, synthetic or natural organic fibers	Sol-gel	SiO ₂	-	P	[50]

*A: article; P: patent

**CVD: chemical vapor deposition

4.4. Photocatalyst Preparation and Immobilization Methods

The preparation of nanometer photocatalyst and immobilization of them can be achieved in one step or in two independent steps one after the other. All methods are summarized in Fig. (1). The film thickness of the immobilized photocatalyst can be influenced during the immobilization step. The optimal film thickness depends on the wavelength of the irradiation source. Short wavelength light (254 nm) is strongly absorbed by the TiO₂ layer, whereas light with longer wavelengths (365 nm) penetrates even thick TiO₂ layers (1.4 μ m) with low energy loss [45].

At first, the synthesis methods without immobilization are considered. Mo *et al.* give a detailed summary of nanosized TiO₂ synthesis [6]. They distinguish between gas phase and liquid phase methods. Gas phase methods are: gas-phase flame hydrolysis, gas-phase oxidation and gas-phase hydrolysis of titanium alkoxides. The liquid-phase methods are: sol-gel method, liquid-phase hydrolysis of TiCl₄, hydrothermal method and water-in-oil microemulsions. The patent of Hugener-Campbell *et al.* [35] describes in detail the sol-gel method. They found that the purity of organometallic precursor is not critical in the synthesis of TiO₂ with a controlled pore distribution (100 to 130 m²/g). Purity of 97% to 99.999% has not shown any differences. The rate of precursor addition affects the reaction in such a way that the faster the addition the faster the hydrolysis is. The best addition rate for isopropoxide is given with 4 drops/5 sec. The hydrolysis rate is also affected by the bulk medium. Non-aqueous, aprotic or non-polar solvents do not

participate in hydrolysis. With the increasing amount of aqueous, protic or polar compound, the hydrolysis rate increases. The pH also affects the hydrolysis. In acidic ambience the hydrolysis occurs faster. Polymers may affect the hydrolysis, if the pH or viscosity is changed by the polymer. The polymer interacts with the solvent and influences the size and shape of the product. If the polymer attracts to the solvent, large void spaces within the polymer chains are formed. Metal salts influence polymer-solvent-interactions and thus affect the shape, surface area, particle size and pore size distribution of the product. If cations are incorporated in vacancies of TiO₂, the band gap is changed. The optimum aging time is given between 72 hours and 168 hours. If the aging time is lower than 72 hours the material has a lower surface area than 100 m²/g. For aging time over 168 hours no further increase in the surface area was reported. Hugener-Campbell *et al.* [35] isolate the solid from the liquid sol after hydrolysis and clean it from residual water and other impurities. They accomplish the isolation by vacuum filtration, gravity filtration, centrifugation or rotoevaporation, whereas the solid will alter under reduced pressure. The isolated solid is refluxed in a solvent that has a lower surface tension than water trapping water and water-based impurities. The reflux solvent is added in excess of water and the reflux time is proportional to the amount of water. The reflux solvent is removed by distillation or an additional drying step. Subsequently, the solid is calcinated at temperatures up to 600 °C producing anatase phase.

If doped or modified photocatalyst is synthesized, the following methods exist: co-precipitation [21, 55], deposition precipitation [21, 55], liquid phase grafting [21, 55-57],

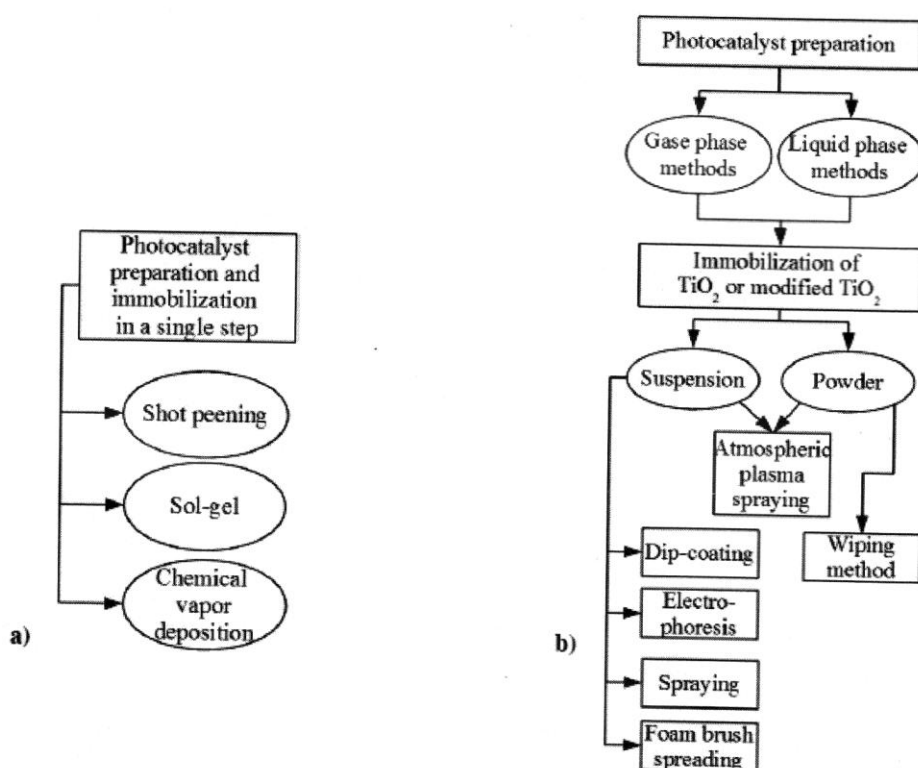


Fig. (1). Photocatalyst preparation and immobilization methods: (a) single, and (b) multistep methods.

colloidal mixing [23] and impregnation [16]. In the co-precipitation method, the precursors of the desired compounds are dissolved and mixed together. Subsequently, the solution is let to precipitate. The precipitate can be filtered and washed thoroughly with distilled or deionized water removing undesired ions. Then the precipitate is dried and calcinated.

In the deposition precipitation method, the photocatalyst powder is dispersed in a solution and added to the dissolved precursor of a doping agent. In distinction to impregnation, a chemical reaction takes place, so that the doping agent precipitates onto the nano-powder surface. The achieved sample can be washed and is dried and calcinated. The advantage of this method is that the doping components are not buried within the photocatalyst particles [21, 55].

In liquid phase grafting method, noble metal complexes are dissolved in an organic solvent, like acetone. This solution reacts with the support surface. Species are formed, which are converted to catalytically active sides during the calcination. Yuan *et al.* [56] and Haruta [57] report that freshly prepared metal hydroxides are needed as support. It is assumed that a high concentration of surface OH-groups are required for the noble metal complex interaction. Whereas, Wei *et al.* add the organic complex solution directly to TiO_2 [21,55]. Later on, the obtained samples are dried and calcinated.

In the impregnation method, precursor salts of the dopant are dissolved in water impregnating the nanoparticles surface. The dopant converts in its catalytic active form during the calcination step.

Methods, where photocatalyst synthesis and immobilization are achieved in a single step, are: shot peening, sol-gel and chemical vapor deposition (CVD). Shot peening was invented by Yoshio and Kasugai-shi [14]. The support surface is preliminary treated by a blasting device. Metal powder is shot by compressed air with high speed onto the metal and/or ceramic support. The powder should have a particle size below 200 μm (preferable 30 – 100 μm). And the pressure should be more than 0.3 MPa, preferably more than 0.5 MPa. The metal powder oxidizes at the surface of the support and forms directly a photocatalytical active layer, which is activated by irradiation with wavelengths longer than UV. The layer exhibits an oxygen deficit tilting structure. Oxygen bonds decrease gradually from the surface to the inside. Yoshio and Kasugai-shi [14] mention the following advantages of the method: no calcination is required, timesaving method in comparison to sol-gel, no binder is required and the layer exhibits good hardness.

Sol-gel method is prevalently encountered in laboratory experiments [1, 33, 44-46, 58]. Typical precursors are metal alkoxides (e.g. tetrabutyl orthotitanate, titanium tetra-isopropoxide), oxysulfates or metal chlorides, which hydrolyse and condensate in an appropriate solvent. The precursor/solvent solution is directly applied to the surface and subsequently dried and calcinated. Kalousek *et al.* [1] added the tri-block-copolymer Pluronic 123® (BASF) to the precursor solution for better dispersing. They calcinated their sample at 350 °C for 3 h and achieved a purely mesoporous film with large surface area and more or less regularly ordered

pores, whose size reaches about 6 -7 nm and whose pore walls are composed of small anatase nanocrystals (about 40%–60%) and some amorphous phase. The sol-gel coated cementitious materials by Ramirez *et al.* [33] showed no significant toluene degradation. They attributed it to the reduced active surface and to the presence of ionic species.

CVD is mentioned in patents as a method for applying photocatalysts onto supports, but they have not clarified the method [13, 21, 23, 55]. In CVD, one or more volatile precursors are deposited on the support surface, where they react and decompose. Plurality of variations exist, like low pressure CVD, atmospheric pressure CVD [40], chemical vapor infiltration [39, 53] and plasma CVD [59]. The morphology and catalytic activity of the deposited film are influenced by the following parameters: pressure, temperature, precursors, vapor fraction, vapor loading, deposition time and substrate material and surface thereof. At low pressure (0.133 kPa) and low temperature (300 °C) fibrous materials are well infiltrated [39, 53]. The temperature influences the crystal phase and diffusion processes. At 500 °C or higher rutile occurs [39, 53]. For stainless steel as well as glass supports, diffusion of ions and atoms was observed for high temperatures which reduced the photocatalytic activity [39, 40]. Evans and Sheel reported that the kind of precursor and support influences the deposited phase [40]. Titanium tetra-isopropoxide formed anatase phase on stainless steel, whereas TiCl_4 formed rutile phase. If SiO_2 was deposited prior to the stainless steel substrate, anatase was formed regardless of precursor and the diffusion of Fe and Cr atoms was inhibited with increasing silica layer.

Immobilization procedures with prepared photocatalyst nanopowders are mainly based on suspensions. The nanopowder is either dispersed in water [12, 27, 30, 37] or organic solvents [35], like ethanol [33] or isopropanol [47]. The suspension thereof contains 1-25 wt% solids. Before the suspension is applied onto the support by the subsequently mentioned procedures, it is recommended to homogenize them. Wei *et al.* [21, 25] suggest in their patents the usage of a homogenizer with a dispersing generator at the speed of 7500 rpm. Agglomerates may also be destructed by sonication (up to 1 h) [30] or centrifugal mixing with an appropriate milling media [35]. For instance, the performance of the tungsten oxide/titanium oxide photocatalyst is enhanced by homogenization [25]. Agglomerates are broken into smaller particles by shear forces, so that the photocatalyst is highly dispensed on the support. Thus, only a single dip or spray process is required to achieve the minimum loading [25]. Furthermore, Wei *et al.* report that the adhesion of photocatalyst is increased by homogenization [21]. Subsequently, the suspension is applied to the surface. In patents dip-coating, electrophoresis and spraying are commonly itemized, whereas the inventors have not further specified the immobilization procedures [13, 21, 25, 29, 35, 55]. Goswami [60] coats the suspension with a foam brush to the support. Dip-coating is a favored procedure mentioned in articles [12, 24, 26-28, 30, 33, 37]. In electrophoresis, the nanoparticles are electrically charged and deposited on the support by an electrical field. If the spraying technique is used, the photocatalyst is deposited onto the support by the kinetic energy of a fluid. A special kind of spray coating is the atmospheric plasma spraying technique. Toma *et al.* [41] and Chen and

Liu [34] used a Plasma –Technik 2000 console and a F4 torch (Sulzer-Metco). The solution feed rates were fixed at 14 g min^{-1} . The plasma spray gun works with dry powders or suspensions. The coating material is injected into the plasma plume, where it is melted and accelerated. The gases forming the plasma plume are mixtures of argon, nitrogen, hydrogen and helium. Hydrogen increases the enthalpy and thermal conductivity of the plasma jet [41]. Argon limits the heat flux due to its low thermal conductivity and helium increases the viscosity. Toma *et al.* [41] investigated plasma sprayed powders and suspensions and they find that plasma sprayed aqueous suspensions exhibit higher photocatalytic activity. The TiO_2 is exposed to reduced temperature since the water has to be evaporated. Thus, the original anatase phase retains and the increase of crystallite size is prevented.

Matsui and Tagata [49] invented a wiping method for coating indoor and outdoor materials with thin photocatalyst films. At first, the surface is cleaned with a cleaning wiping material. The photocatalyst is applied by unwoven cloth of super –extra fine chemical fibers, which are soaked in the photocatalyst formulation. Excess formulation is wiped off by wiping material. The composition of formulation is given in Table 5. The photocatalyst concentration is between 0.1 wt%-20 wt%, so that enough catalyst is spread and that the transparency of the surface is retained. The ratio of a binder agent is preferably 0.1 wt%-20 wt%. For lower concentration than 0.1 wt% the adherence of photocatalyst is not strong enough. Binder concentration above 20 wt% decreases the fluidity. The auxiliary binder spreads the photocatalyst and prevents agglomeration. If the auxiliary binder exceeds 70 wt% the photocatalyst layer hardens and cannot be spread. The dispersion agent stabilizes the suspension. The abrasive agent wipes off the unnecessary binders, it

should not exceed 50 wt% since the binder agent would not be sufficient. Antioxidant agents can be added to prevent discoloration which originates in the oxidation of auxiliary binders.

5. PHOTOREACTORS FOR GAS PHASE TREATMENT/PURIFICATION

Photocatalytic reactors are designed in order to provide strictly controlled conditions for photocatalytic air purification. Some reactors can be utilized in commercial applications as a part of HVAC systems (Heating, Ventilation and Air Conditioning) but most of them are used in laboratories to measure the activity of different types of photocatalysts used for gas streams treatment. Many types of photoreactors are designed. Plate, annular, honeycomb-monolith, packed-bed and fluidized-bed reactors are the most popular, but in the literature other types are also described: with aerosol generator, with optical fibers, with TiO_2 -coated foam nickel, hybrid non-thermal plasma reactors and other. As a typical source of irradiation there are used different kinds of UV lamps, but some researchers also use visible light [19, 26]. To maximize the efficiency of photocatalytic reactions Taranto *et al.* presented the methodical strategy for modeling and optimizing irradiance on some kinds of photocatalyst [61].

5.1. Plate Reactors

A plate reactor, shown in Fig. (2), is the simplest type of photoreactor used for gas phase treatment. The typical form of this reactor is a rectangular box, made of plexiglass [33], polycarbonate [41], stainless steel [19] or other material, that

Table 5. Formulation of Photocatalyst Applied by Wiping Method [49]

Compound	Chemicals	Examples	Amount
Photocatalyst	-	Pt- TiO_2	0.1 wt%-20 wt%
Binder	Resins, SiO_2	Silicon resin, acrylic fiber resin, flourine resin, epoxy resin	0.1 wt%-20 wt%
Auxiliary binder	Mineral spirit	Mineral turpentine, white spirit, mineral thinner, petroleum spirit	20 wt%-70 wt%
	Vegetable oil	Camellia oil, olive oil, sesame oil, soybean oil, safflower oil, colza oil, evening primrose oil	
	Wax	Calbana-wax, candy-wax, comenuca-wax, ouricuri-wax, sugarcane wax, hohoba-wax, abracate seed-wax, moc-wax, japan-wax, propolis	
Dispersing agent	Amino acids	Teanin, lysine, glutamic acid, aspartic acid, taurine	0.01 wt%-1 wt%
	Saccharose fatty acid esters	Saccharose stearic acid esters, saccharose palmitin acid esters	
	Polyphenols	Catechin	
Antioxidant agent	-	Tilhidorokishitolen, tocoferol, fitin sannad, fenotiagen, getiohosfat, getiocalbat, etc.	not essential
Antifreeze agent	-	Ethylen glycol	not essential
Abrasive agent	Ceramic material (30-50 nm)	SiO_2	0.1 wt%-50 wt%
Organic solvent	-	Ethanol, butyl acetat, cyclohexane, propylen glycol methylic ether, propylen glycol methylic ether acetat	the remainder

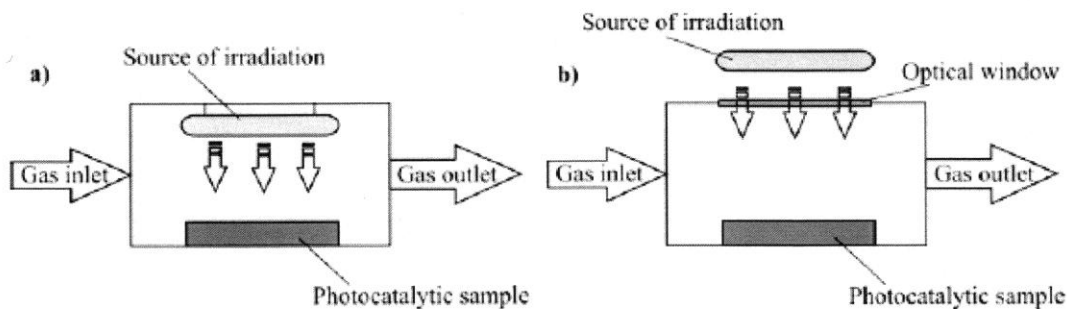


Fig. (2). A typical arrangement of plate reactors with inner (a) and external (b) source of irradiation.

is non adsorbing to applied pollutants and resistant to UV radiation [32]. In some applications, the surface of the reactor is coated by a Teflon film [12, 27, 37]. Testing samples used in plate reactors are typically planar or have a form of powders and they are placed at the bottom of the reactor.

Plate photoreactors can be divided into two groups: with inner and with outer source of irradiation. In the first group (Fig. 2a), a lamp is placed at the upper part of the reactor and there is no separation between the lamp and the testing sample [12, 27, 37, 62]. In the second group (Fig. 2b), reactors are equipped with a window, which permits the light passage from irradiation source (placed above the reactor) into the testing sample [41]. This window can be made of borosilicate [17], plexiglas [41], quartz [54] or other material allowing the UV or visible light to pass through. Some reactors with outer source of irradiation are equipped with seals and profiles so that the sample gas only passes the reactor through the slit between the sample surface and the glass cover in longitudinal direction (Fig. 3). Structural parts inside the reactor can be designed to enable laminar flow and to avoid turbulences [17, 32].

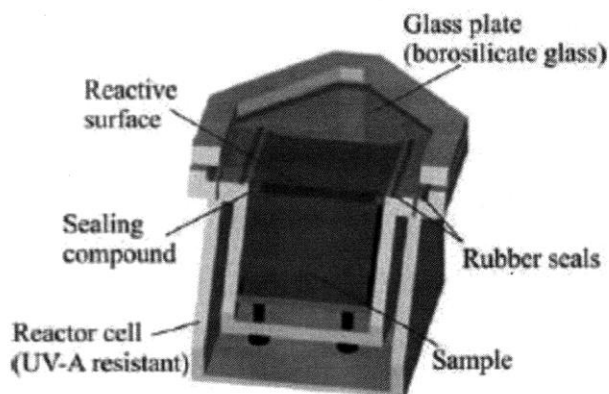


Fig. (3). Photocatalytic reactor described in [17] and [32].

Another type of a plate reactor with outer source of light was examined by Leung *et al.* [58]. This reactor consists of borosilicate, transparent plates, coated with TiO_2 thin films. The plates are packed parallel inside the reactor, resulting in a stack of multiple modular reactors (Fig. 4).

Typical plate photoreactors are not used in commercial systems for air purification, but they are frequently utilized

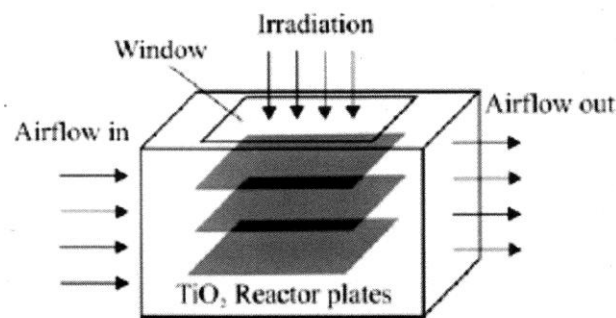


Fig. (4). Photoreactor used by Leung *et al.* [58].

in laboratories to study the photoactivity of different kinds of photocatalysts. As model air pollutants there are used VOCs (Volatile Organic Compounds), nitric oxides (NO_x), carbon monoxide (CO) and others. More details are compiled in Table 6.

5.2. Annular-flow Photocatalytic Reactors

Fig. (5) shows a scheme of typical annular-flow photocatalytic reactor. This reactor consists of a cylindrical tube mostly made of Pyrex glass with the inner surface coated with the photocatalyst. Longitudinal light source (UV lamp) is located in the center of the glass tube. The airflow is provided along the axial direction through the annulus between the lamp and the tube [44].

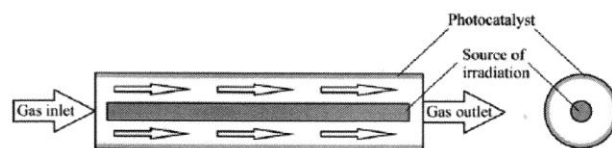


Fig. (5). Typical annular-flow photocatalytic reactor [44].

Shiraishi *et al.* [42, 43, 63] used the photocatalytic reactor with a parallel array of nine light sources. Blacklight fluorescent lamps were placed in the center of the glass tubes with the inner surface coated with a thin film of TiO_2 . These nine reaction units were equipped with an electric fan providing the airflow and everything was placed inside a plastic box (Fig. 6). The distance between the UV lamp and the

Table 6. Plate Photoreactors Used for Gas Phase Treatment

Type of Plate Reactor	Material of the Reactor	Irradiation Source	Photocatalyst	Applications/Measurements	Scheme	Ref.
With inner source of irradiation	Material not claimed. Reactor's surface coated by a Teflon film	6W UV lamp (Cole-Parmer) - primary wavelength 365 nm	TiO ₂ - coated glass fiber filter (Whatman)	Effect of NO on the photodegradation of CO and NO ₂	No scheme	[27]
	Material not claimed. Reactor's surface coated by a Teflon film	6W UV lamp (Cole-Parmer) - primary wavelength 365 nm	TiO ₂ - coated glass fiber filter (Whatman)	Removal of VOCs and NO ₂	Schematic diagram of the experimental setup	[12]
	Material not claimed. Reactor's surface coated by a Teflon film	6W UV lamp (Cole-Parmer) - primary wavelength 365 nm	TiO ₂ - coated glass fiber filter (Whatman)	Inhibition effect of SO ₂ on NO _x and VOCs photodegradation	Schematic diagram of the experimental setup	[37]
	Stainless steel reactor with the equipment (gas compartment, illuminator, ventilator, tray support, admission valve, discharge valve, humidifier, sensors)	Two kinds of UV lamps: with primary wavelength of 365 nm and with primary wavelength of 254 nm	TiO ₂ immobilized on the surface of asphalt material	Heterogeneous photocatalysis for NO _x removal from vehicle emissions	Schematic cross section of the reactor and its equipment	[8]
With outer source of irradiation	Non-adsorbing plastic material. Top of the reactor tightly covered with a glass plate (borosilicate)	25W cool day light lamps (Philips, the Netherlands) - visible radiation in the range of 400-700 nm	Carbon-doped TiO ₂ on gypsum plasterboard cover paper with non-woven glass wool fabric (wallpaper)	Removal of NO (as typical indoor air pollutant).	Schematic diagram of the reactor. Schematic diagram of the experimental setup	[17]
	Material not claimed (non adsorbing to applied pollutant and UV-A radiation-resistant). Top of the reactor tightly covered with a glass plate (borosilicate)	25W fluorescent tubes, emitting a high-concentrated UV-A radiation in the range of 300-400 nm (maximum intensity at about 345 nm)	Photocatalytic concrete products (paving blocks)	Removal of NO	Schematic diagram of the reactor. Schematic diagram of the experimental setup	[32]
	Plexiglass reactor closed by borofloat glass-plate	18W blacklight blue UV lamp (Philips Lighting, Turnhout, Belgium) - radiation range 340-410 nm (maximum emission at 365 nm)	TiO ₂ in concrete and plaster materials used for wall and floor covering. TiO ₂ on autoclaved aerated white concrete and concrete tiles.	Removal of toluene	Schematic diagram of the experimental setup	[33]
	Polycarbonate box with a Plexiglas window placed in environmental chamber.	UV lamp with a fraction of 30% UVA and 4% UVB placed inside the environmental chamber, above the reactor	Steel plates with thin films of TiO ₂ , deposited by conventional plasma spraying and liquid plasma spraying	Removal of NO _x	Schematic diagram of the experimental setup	[41]
	Stainless steel reactor covered with Saint-Glass	300W commercial tungsten halogen lamp (General Electric) - simulates solar light	Dishes containing the photocatalysts powders (C-N-S-tridoped TiO ₂)	Removal of NO	No scheme	[19]

(Table 6) Contd....

Type of Plate Reactor	Material of the Reactor	Irradiation Source	Photocatalyst	Applications/Measurements	Scheme	Ref.
	Stainless steel reactor with a quartz window	200-W mercury lamp	Glass plate coated with surface-fluorinated TiO_2	Photocatalytic oxidation of gaseous acetaldehyde	Schematic diagram of the experimental setup	[54]
	Material not claimed.	5 W UVA lamp	Borosilicate glass plates coated with sol-gel TiO_2 thin films on both sides. Some plates are packed parallel inside the photoreactor	Photocatalytic degradation of formaldehyde vapor	Schematic diagram of the reactor. Schematic diagram of the experimental setup	[58]

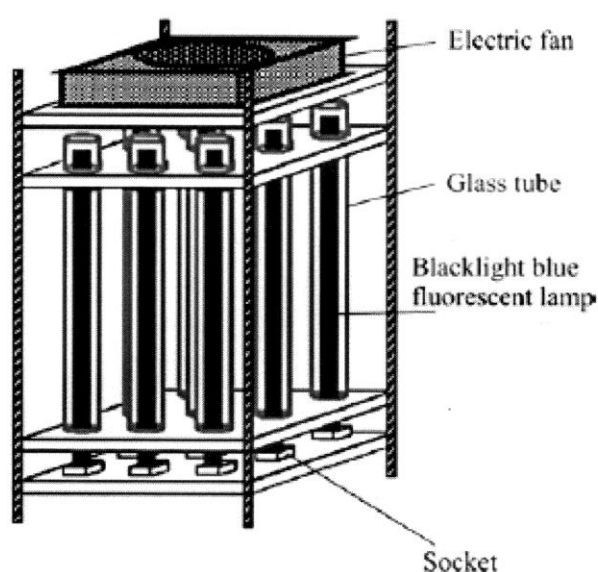


Fig. (6). Annular flow reactor used by Shiraishi *et al.* [43].

photocatalyst was small (5mm), which provided high illumination intensity and high linear velocity of polluted air passing through the reactor. The consequence of high linear velocity is that there is no diffusional resistance during the photocatalytic reaction. In addition, the UV radiation permeates through the glass and TiO_2 film into the catalyst of the neighboring tubes increasing reaction rate [42]. Thus, this kind of photoreactors, equipped with adsorption/desorption unit, provides very good efficiency in removal of formaldehyde [43] and toluene [42] from indoor air. Another practical application for this type of reactors is claimed in [52], where photocatalytic reactors were used for disinfection of air.

Pichat *et al.* [38] developed another type of photoreactor, which is similar to annular-flow reactors (total airflow is parallel to light source), but the photocatalyst is not applied on the surface of a glass tube (Fig. 7). UV lamp is surrounded by some layers of the Matrix material (TiO_2 -coated fiber glass mesh). The gas flow in this reactor is turbulent

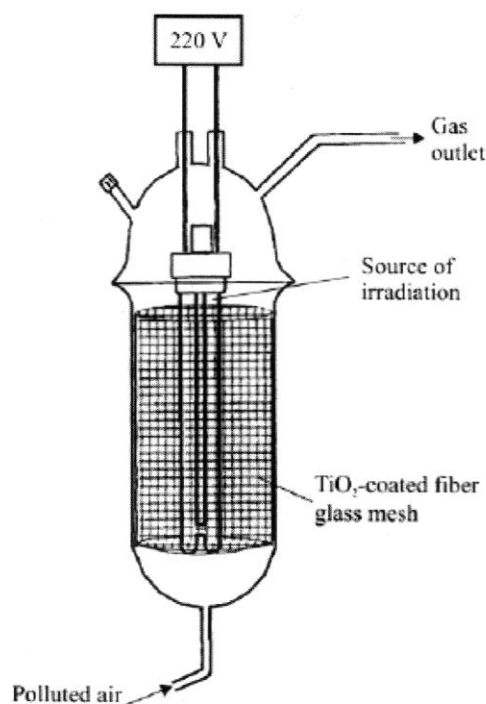


Fig. (7). Photocatalytic reactor with TiO_2 -coated fiber glass mesh used by Pichat *et al.* [38].

and the radiation is completely adsorbed/diffracted by the catalytic material.

5.3. Honeycomb Monolith Reactors

Honeycomb monolith reactors are very popular and widely used for thermally activated reactions (for example catalytic converters in cars), but they also can be used for photocatalytic reactions. This type of reactors contains of a certain number of parallel channels of circular, square [47] or hexagonal [30] cross section. For the photocatalytic applications, the inner walls of the channels are coated with a thin film of photocatalyst. External irradiation source is located in front of the channels. The radiation field modeling in a

photocatalytic monolith reactor was described by Hossain *et al.* [47].

Taranto *et al.* [30] used an aluminum honeycomb material with hexagonal cells, coated with a thin film of TiO_2 . As a source of light, low-pressure mercury lamps were used. The scheme of the photocatalytic reactor they used is shown in Fig. (8).

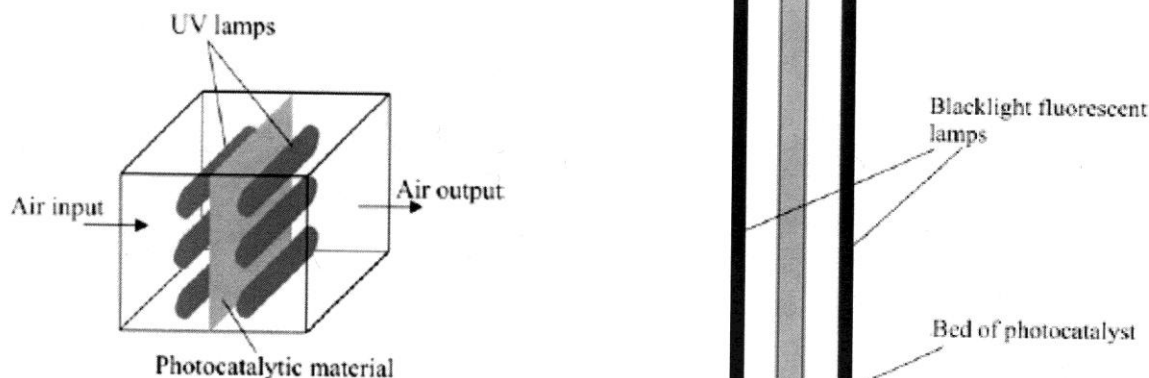


Fig. (8). Honeycomb monolith reactor used by Taranto *et al.* [30].

Photocatalytic reactors based on honeycomb monoliths yield lower pressure drop than other types of photocatalytic reactors used for air purification [47]. For this reason, some air purification systems based on honeycomb monoliths were developed. They are described elsewhere [21, 25, 36].

5.4. Packed-Bed Reactors

Obuchi *et al.* [22] used packed-bed reactor to measure the photocatalytic properties of TiO_2 on porous silica ($\text{TiO}_2/\text{SiO}_2$) in removal of acetaldehyde from air. The reactor (Fig. 9) consisted of a Pyrex glass tube. Catalyst particles were placed inside the tube and four light sources (blacklight fluorescent lamps) were located outside. Polluted air was supplied to the upper part of the packed bed.

5.5. Fluidized-Bed Reactors

The photocatalyst in fluidized-bed reactor is effectively exposed to UV radiation and has a good contact with the stream of treated gas. Dibble and Raupp [64] used this type of a reactor to oxidize trichloroethylene in contaminated airstreams. The scheme of the flat-wall reactor (made of Pyrex glass) they used is shown in Fig. (10). Coarse glass frits were placed in the gas inlet and outlet to keep the catalyst bed (TiO_2 on silica gel) inside the reactor. UV irradiation was provided by a single fluorescent UV lamp located parallel to the reactor's wall.

Prieto *et al.* [65] used the fluidized-bed reactor to remove toluene from air. The cylindrical reactor (Fig. 11) was made of Pyrex glass and the irradiation was provided by a low-pressure Hg vapor UV lamp placed outside the reactor. As the photocatalyst there were used nonporous and nonspherical particles, produced by the calcination of the mixture of TiO_2 with starch in distilled water and a methanol solution.

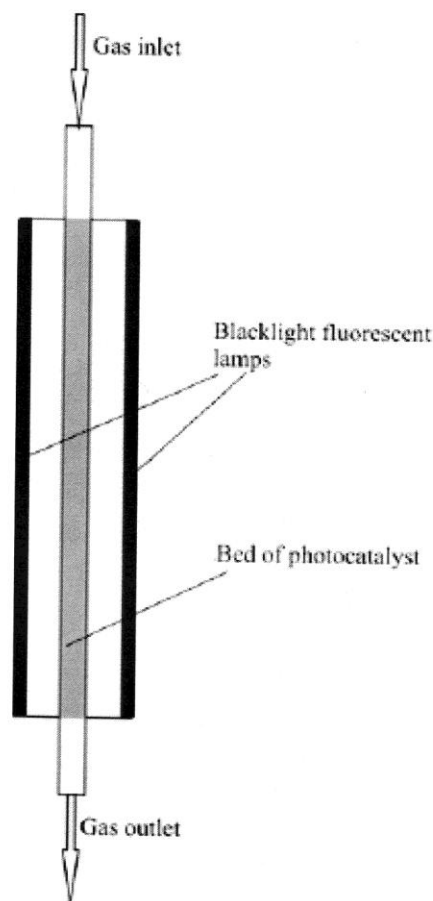


Fig. (9). Packed-bed reactor described by Obuchi *et al.* [22].

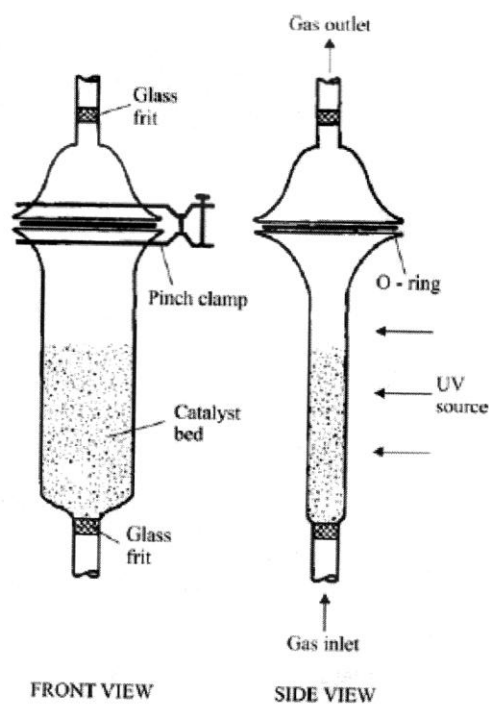


Fig. (10). Fluidized-bed reactor used by Dibble and Raupp [64].

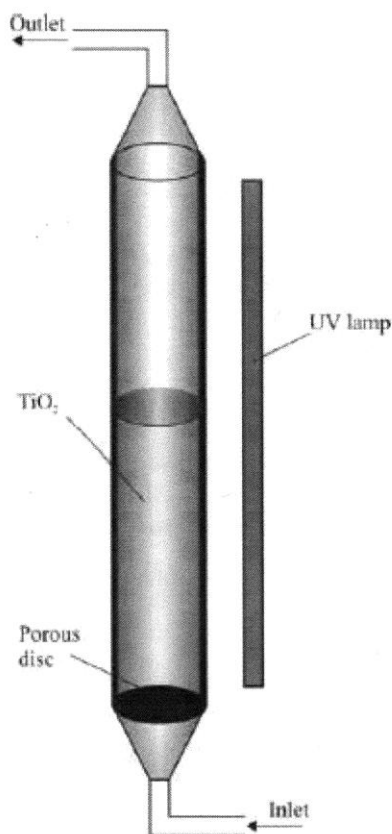


Fig. (11). Fluidized-bed reactor used by Prieto *et al.* [65].

5.6. Reactors with Aerosol Generator

Photocatalytic reactor with the generator of the TiO_2 aerosol was described by Besov *et al.* [66] and it was used to remove acetone and dimethyl methylphosphonate from air. In a cubic chamber made of Plexiglas (Fig. 12), there was placed an annular UV fluorescent Hg light and a sonic generator of TiO_2 aerosol. The basic part of this generator was a small speaker, connected to a signal generator (200 Hz) and located inside a plastic cylinder. TiO_2 powder was placed on the surface of the speaker's membrane. After turning-on the electrical power, the catalyst was sprayed into the chamber from a small orifice in the plastic cylinder.

5.7. Reactors with Optical Fibers

Hager *et al.* [67] used the photoreactor with catalyst-coated optical fibers to study the efficiency of degradation of toluene, 2-propanol and acetone in the stream of air. This novel type of photoreactor was described by Hofstadler *et al.* [68] and this research group used it to wastewater treatment. Fig. (13) shows a scheme of the reactor, which consisted of a glass tube with a packet of 140 TiO_2 -coated fused-silica glass fibers, supported by perforated Teflon plates. In front of the ends of the fibers there was placed an UV transparent

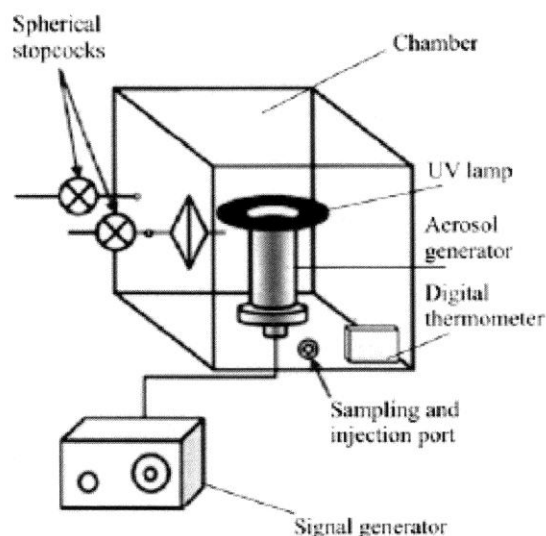


Fig. (12). Reactor with the generator of TiO_2 aerosol [66].

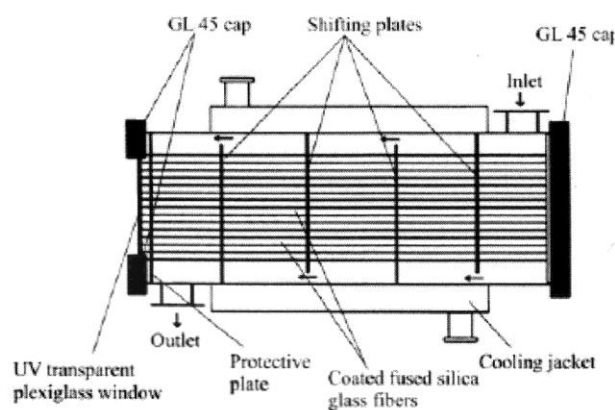


Fig. (13). Photoreactor with catalyst-coated optical fibers [68].

Plexiglass window. Behind the window, a Teflon protective plate was placed to avoid direct photolysis of treated contamination. As a source of UV irradiation, a lamp equipped with a 3-in. fused-silica condenser and 400W Hg lamp were used.

5.8. Reactors with TiO_2 -Coated Foam Nickel

Yang *et al.* [45, 46] described a novel type of a photoreactor for photocatalytic oxidation of gaseous formaldehyde which is shown in Fig. (14). This reactor contained 15 parallel placed reaction cells made of a TiO_2 -coated tubular foam nickel with a UV lamp placed in the center. One end of each reaction cell was sealed in order to provide axial and then radial direction of air stream. In [45] the system for air purification equipped only with the photocatalytic reactor was described and in [46] the research group used hybrid system containing the photocatalytic reactor and supporting VUV lamp placed in front of it.

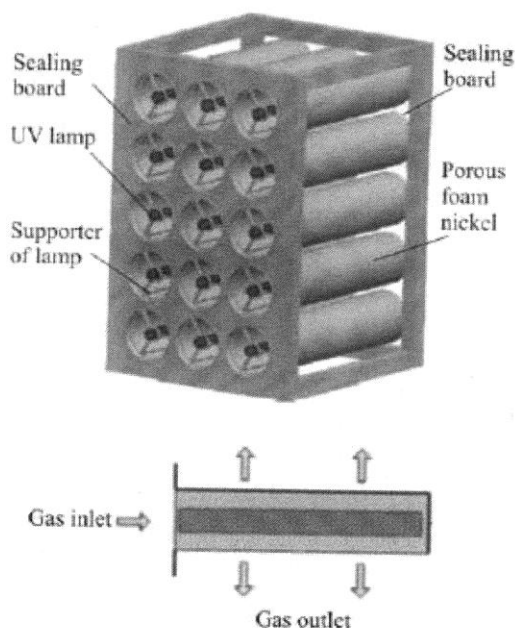


Fig. (14). Reactor with TiO_2 -coated foam nickel [45, 46].

5.9. Non-thermal Plasma Reactors

In this type of photoreactors, gaseous pollutants are treated simultaneously by cold plasma and by the action of the photocatalyst under UV irradiation. An example of this type of reactors is shown in Fig. (15). An active part of the reactor (made of Pyrex glass) was rectangular, and it was equipped with two copper electrodes (placed outside). The electrodes were connected to a generator, providing sinusoidal voltage of 30 kV at a frequency of 50 Hz. This high voltage was needed to generate plasma. The photocatalyst, made of fiberglass weaving coated with colloidal silica and TiO_2 was placed inside the active part of the reactor. Four UV lamps, providing additional UV irradiation were located

around the photoreactor, and acetylene was used as air pollutant. More technical details are described in [50].

Nasonova *et al.* [10] used non-thermal plasma reactor without outer source of UV irradiation, because plasma was utilized as the source of UV light. The scheme of this reactor (used for NO and SO_2 removal) is shown in Fig. (16). Glass beads coated with TiO_2 thin film, used as photocatalyst, were located inside a glass tube. The first electrode (copper rod wire) was placed at the center of the reactor and the second one (stainless steel mesh) surrounded the tube.

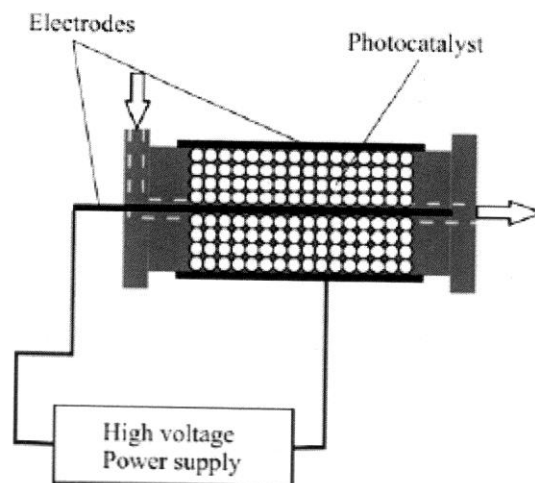


Fig. (16). Non-thermal plasma reactor described by Nasonova *et al.* [10].

6. CURRENT & FUTURE DEVELOPMENTS

The photocatalytic oxidation of organic compounds in aqueous environment has received the most attention, but there is a rapidly increasing focus on the oxidation of volatile organic or inorganic compounds in the gas phase, including VOC, NO_x and SO_x . To improve gas phase treatments effi-

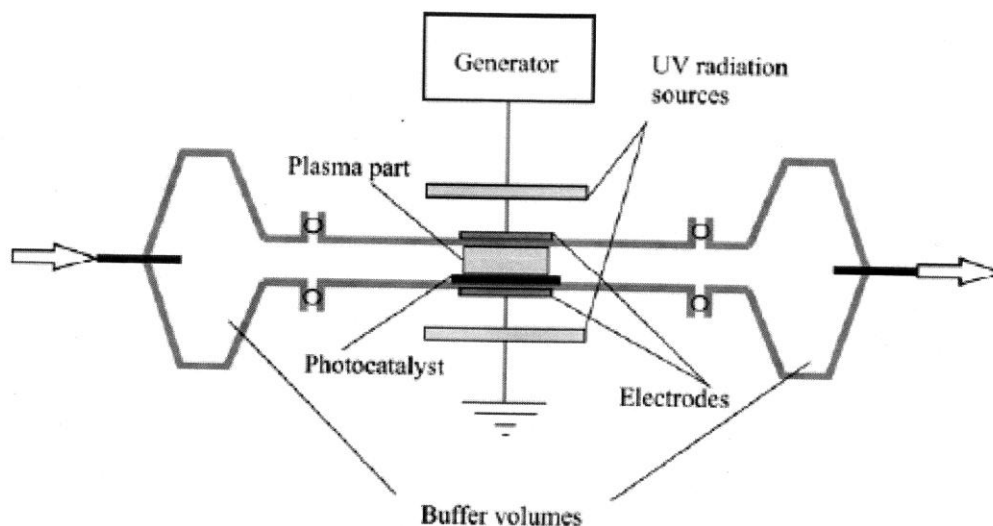


Fig. (15). Non-thermal plasma reactor described in [50].

ciency two main approaches could be taken into consideration: (a) TiO_2 modification to increase the spectral sensitivity of TiO_2 -based photocatalysts to visible light and to compete with electron-hole recombination (by metal and non-metal doping or co-doping), and (2) enhancement of interface surface area and light utilization by photoreactor design.

TiO_2 -based photocatalysts, used for gas phase treatment/purification, usually are prepared by gas phase methods (e.g. gas-phase flame hydrolysis, gas-phase oxidation and gas-phase hydrolysis of titanium alkoxides) or the liquid-phase methods (e.g. sol-gel method, liquid-phase hydrolysis of TiCl_4 , hydrothermal method and water-in-oil microemulsions). To enhance interface surface area and light utilization different types of photoreactors are designed, such as fluidized-bed reactor, reactor with TiO_2 aerosol generator or equipped with TiO_2 -coated optical fibers. Still one of the major challenges for the scientific and industrial community involved in photocatalytic research is to increase the spectral sensitivity of TiO_2 -based photocatalysts to visible light to utilize solar light as a cheap source of renewable energy. Therefore, future research and patents would deal with visible light-activated photocatalyst and photoreactors functioning in the presence of solar irradiation.

CONFLICT OF INTEREST

None.

REFERENCES

- [1] Kalousek V, Tschirch J, Bahnemann D, Rathousk J. Mesoporous layers of TiO_2 as highly efficient photocatalysts for the purification of air. *Superlattice Microstat* 2008; 44: 506-13.
- [2] Fujishima A, Rao TN, Tryk DA. Titanium dioxide photocatalysis. *J Photochem Photobiol C* 2000; 1: 1-21.
- [3] Fujishima A, Zhang X. Titanium dioxide photocatalysis: present situation and future approaches. *Cr Chim* 2006; 9: 750-60.
- [4] Hoffmann MR, Martin ST, Choi W, Bahnemann DW. Environmental applications of semiconductor photocatalysis. *Chem Rev* 1995; 95: 69-96.
- [5] Zhao J, Yang X. Photocatalytic oxidation for indoor air purification: a literature review. *Build Environ* 2003; 38: 645-54.
- [6] Mo J, Zhang Y, Xu Q, Lamson JJ, Zhao R. Photocatalytic purification of volatile organic compounds in indoor air: A literature review. *Atmos Environ* 2009; 43: 2229-46.
- [7] Feiyan C, Pehkonen SO, Ray MB. Kinetics and mechanisms of UV-photodegradation of chlorinated organics in the gas phase. *Water Res* 2002; 36: 4203-14.
- [8] Chen M, Liu Y. NO_x removal from vehicle emissions by functionality surface of asphalt road. *J Hazard Mater* 2010; 174: 375-9.
- [9] Lim TH, Jeong SM, Kim SD, Gyenis J. Degradation characteristics of NO by photocatalysis with TiO_2 and CuO/TiO_2 . *React Kinet Catal L* 2000; 71: 223-9.
- [10] Nasonova A, Pham HC, Kim DJ, Kim KS. NO and SO_2 removal in non-thermal plasma reactor packed with glass beads- TiO_2 thin film coated by PCVD process. *Chem Eng J* 2010; 156: 557-61.
- [11] Maggos T, Bartzis JG, Liakou M, Gobin C. Photocatalytic degradation of NO_x gases using TiO_2 -containing paint: A real scale study. *J Hazard Mater* 2007; 146: 668-73.
- [12] Ao CH, Lee SC, Yu JC. Photocatalyst TiO_2 supported on glass fiber for indoor air purification: effect of NO on the photodegradation of CO and NO_2 . *J Photochem Photobiol A* 2003; 156: 171-7.
- [13] Wei D, Radhakrishnan R, Vanderspurt TH. Bifunctional manganese oxide/titanium dioxide photocatalyst/thermocatalyst. WO2004112941, 2004.
- [14] Yoshio M, Kasugai-shi A. A metal oxide layer prepared by shot peening. EP1192994, 2002.
- [15] Han F, Kambala VSR, Srinivasan M, Rajarathnam D, Naidu R. Tailored titanium dioxide photocatalysts for the degradation of organic dyes in wastewater treatment: A review. *Appl Catal A-Gen* 2009; 359: 25-40.
- [16] Kitaguchi S. Visible light-responsive photocatalyst composition and process for producing the same. US2008119352, 2008.
- [17] Yu QL, Brouwers HJH. Indoor air purification using heterogeneous photocatalytic oxidation part I: experimental study. *Appl Catal B-Environ* 2009; 92: 454-61.
- [18] Xiao G, Wang X, Li D, Fu X. In VO_4 -sensitized TiO_2 photocatalysts for efficient air purification with visible light. *J Photochem Photobiol A* 2008; 193: 213-21.
- [19] Wang Y, Huang Y, Ho W, Zhang L, Zou Z, Lee S. Biomolecule-controlled hydrothermal synthesis of CNS-tridoped TiO_2 nanocrystalline photocatalysts for NO removal under simulated solar light irradiation. *J Hazard Mater* 2009; 169: 77-87.
- [20] Yu YT. TiO_2 -capsulated metallic nanoparticles photocatalyst enable to be excited by UV or visible lights and its preparation method. WO2009048186, 2009.
- [21] Wei D, Obee TN, Hay SO, Vanderspurt TH, Schmidt WR. Bifunctional layered photocatalyst/thermocatalyst for improving indoor air quality. US20050129591, 2005.
- [22] Obuchi E, Sakamoto T, Nakano K, Shiraishi F. Photocatalytic decomposition of acetaldehyde over $\text{TiO}_2/\text{SiO}_2$ catalyst. *Chem Eng Sci* 1999; 54: 1525-30.
- [23] Vanderspurt TH, Davies JA, Hay SO, Obee TN, Opalka SM, Wei D. Air purification system. US20090246091, 2009.
- [24] Shelimov BN, Tolkachev NN, Tkachenko OP, et al. Enhancement effect of TiO_2 dispersion over alumina on the photocatalytic removal of NO_x admixtures from O_2 - N_2 flow. *J Photochem Photobiol A* 2008; 195: 81-8.
- [25] Wei D, Obee TN, Hay SO, Vanderspurt TH, Schmidt WR, Sangiovanni JJ. Tungsten oxide/titanium dioxide photocatalyst for improving indoor air quality. US7255831, 2007.
- [26] Yu BF, Hu ZB, Liu M, Yang HL, Kong QX, Liu YH. Review of research on air-conditioning systems and indoor air quality control for human health. *Int J Refrig* 2009; 32: 3-20.
- [27] Ao CH, Lee SC, Mak CL, Chan LY. Photodegradation of volatile organic compounds (VOCs) and NO for indoor air purification using TiO_2 : promotion versus inhibition effect of NO. *Appl Catal B-Environ* 2003; 42: 119-29.
- [28] Pal A, Pehkonen SO, Yu LE, Ray MB. Photocatalytic inactivation of Gram-positive and Gram-negative bacteria using fluorescent light. *J Photochem Photobiol A* 2007; 186: 335-41.
- [29] Wei D, Vanderspurt TH, Radhakrishnan R, Hay SO, Obee TN, Schmidt WR. Multi-layered photocatalyst/thermocatalyst for improving indoor air quality. US20050129589, 2005.
- [30] Taranto J, Frochot D, Pichat P. Photocatalytic air purification: Comparative efficacy and pressure drop of a TiO_2 -coated thin mesh and a honeycomb monolith at high air velocities using a 0.4 m³ close-loop reactor. *Sep Purif Technol* 2009; 67: 187-93.
- [31] Ehrt D. Structure, properties and applications of borate glasses. *Glass Technol-Part A* 2000; 41: 182-5.
- [32] Hüsken G, Hunger M, Brouwers HJH. Experimental study of photocatalytic concrete products for air purification. *Build Environ* 2009; 44: 2463-74.
- [33] Ramirez AM, Demeestere K, De Belie N, Mäntylä T, Levänen E. Titanium dioxide coated cementitious materials for air purifying purposes: Preparation, characterization and toluene removal potential. *Build Environ* 2009; 45: 832-8.
- [34] Chen M, Liu Y. NO_x removal from vehicle emissions by functionality surface of asphalt road. *J Hazard Mater* 2010; 174: 375-9.
- [35] Hugener-Campbell T, Vanderspurt TH, Schmidt WR, Zhitnik SM. Deactivation resistant photocatalyst and method of preparation. WO2008147359, 2008.
- [36] Garfield K, Potter J. System and method for photocatalytic oxidation air filtration using a substrate with photocatalyst particles power coated thereon. US20080286163, 2008.
- [37] Ao CH, Lee SC, Zou SC, Mak CL. Inhibition effect of SO_2 on NO_x and VOCs during the photodegradation of synchronous indoor air pollutants at parts per billion (ppb) level by TiO_2 . *Appl Catal B-Environ* 2004; 49: 187-93.
- [38] Pichat P, Disdier J, Hoang-Van C, Mas D, Goutailler G, Gaysse C. Purification/deodorization of indoor air and gaseous effluents by TiO_2 photocatalysis. *Catal Today* 2000; 63: 363-9.

- [39] Sarantopoulos C, Puzenat E, Guillard C, Herrmann JM, Gleizes AN, Maury F. Microfibrous TiO₂ supported photocatalysts prepared by metal-organic chemical vapor infiltration for indoor air and waste water purification. *Appl Catal B-Environ* 2009; 91: 225-33.
- [40] Evans P, Sheel DW. Photoactive and antibacterial TiO₂ thin films on stainless steel. *Surf Coat Tech* 2007; 201: 9319-24.
- [41] Toma FL, Bertrand G, Chwa SO, Meunier C, Klein D, Coddet C. Comparative study on the photocatalytic decomposition of nitrogen oxides using TiO₂ coatings prepared by conventional plasma spraying and suspension plasma spraying. *Surf Coat Tech* 2006; 200: 5855-62.
- [42] Shiraishi F, Ishimatsu T. Toluene removal from indoor air using a miniaturized photocatalytic air purifier including a preceding adsorption/desorption unit. *Chem Eng Sci* 2009; 64: 2466-72.
- [43] Shiraishi F, Yamaguchi S, Ohbuchi Y. A rapid treatment of formaldehyde in a highly tight room using a photocatalytic reactor combined with a continuous adsorption and desorption apparatus. *Chem Eng Sci* 2003; 58: 929-34.
- [44] Shiraishi F, Ohkubo D, Toyoda K, Yamaguchi S. Decomposition of gaseous formaldehyde in a photocatalytic reactor with a parallel array of light sources: 1. Fundamental experiment for reactor design. *Chem Eng J* 2005; 114: 153-9.
- [45] Yang L, Liu Z, Shi J, Hu H, Shanguan W. Design consideration of photocatalytic oxidation reactors using TiO₂-coated foam nickels for degrading indoor gaseous formaldehyde. *Catal Today* 2007; 126: 359-68.
- [46] Yang L, Liu Z, Shi J, Zhang Y, Hu H, Shanguan W. Degradation of indoor gaseous formaldehyde by hybrid VUV and TiO₂/UV processes. *Sep Purif Technol* 2007; 54: 204-11.
- [47] Hossain M. Radiation field modeling in a photocatalytic monolith reactor. *Chem Eng Sci* 1998; 53: 3771-80.
- [48] Reisfeld B, Chiang RHL, Josserand O, Dunshee KB, Jomard T, Drago TE. Modular photocatalytic air purifier. US2003019738, 2003.
- [49] Matsui N, Tagata G. Method for decomposition and removal of organic compound in air using platinum as main catalysis, method for formation of photocatalytic article and photocatalytic layer, and photocatalyst. US2009136405, 2009.
- [50] Thevenet F, Puzenat E, Guillard C, et al. Combined treatment of gaseous effluents by cold plasma and photocatalysis. WO2007051912, 2007.
- [51] Muggli DS. Photocatalytic fluidized bed air purifier. WO2007076134, 2007.
- [52] Longstaff DE. Structure and method of air purification. US20090098014, 2009.
- [53] Sarantopoulos C, Gleizes AN, Maury F. Chemical vapor infiltration of photocatalytically active TiO₂ thin films on glass microfibers. *Surf Coat Tech* 2007; 201: 9354-8.
- [54] Kim H, Choi W. Effects of surface fluorination of TiO₂ on photocatalytic oxidation of gaseous acetaldehyde. *Appl Catal B-Environ* 2007; 69: 127-32.
- [55] Wei D, Vanderspurt TH, Obee TN, Hay SO, Schmidt WR. Purification system comprising gold/titanium dioxide photocatalyst. WO2004112958, 2004.
- [56] Yuan Y, Kozlova AP, Asakura K, Wan H, Tsai K, Iwasawa Y. Supported Au catalysts prepared from Au phosphine complexes and as-precipitated metal hydroxides: characterization and low-temperature CO oxidation. *J Catal* 1997; 170: 191-9.
- [57] Haruta M. Nanoparticulate gold catalysts for low-temperature CO oxidation. *J New Mat Elect Syst* 2004; 7: 163-72.
- [58] Leung MKH, Tang SM, Lam RCW, et al. Parallel-plate solar photocatalytic reactor for air purification: Semi-empirical correlation, modeling, and optimization. *Sol Energy* 2006; 80: 949-55.
- [59] Nasonova A, Pham HC, Kim DJ, Kim KS. NO and SO₂ removal in non-thermal plasma reactor packed with glass beads-TiO₂ thin film coated by PCVD process. *Chem Eng J* 2009; 156: 557-61.
- [60] Goswami DY. Photoelectrochemical air disinfection. US2006188388, 2006.
- [61] Taranto J, Frochot D, Pichat P. Modeling and optimizing irradiance on planar, folded, and honeycomb shapes to maximize photocatalytic air purification. *Catal Today* 2007; 122: 66-77.
- [62] Bolashikov ZD, Melikov AK. Methods for air cleaning and protection of building occupants from airborne pathogens. *Build Environ* 2009; 44: 1378-85.
- [63] Shiraishi F, Toyoda K, Miyakawa H. Decomposition of gaseous formaldehyde in a photocatalytic reactor with a parallel array of light sources: 2. Reactor performance. *Chem Eng J* 2005; 114: 145-51.
- [64] Dibble LA, Raupp GB. Fluidized-bed photocatalytic oxidation of trichloroethylene in contaminated air streams. *Environ Sci Technol* 1992; 26: 492-5.
- [65] Prieto O, Feroso J, Irusta R. Photocatalytic degradation of toluene in air using a fluidized bed photoreactor. *Int J Photoenergy* 2007; 9: 32859.
- [66] Besov AS, Vorontsov AV, Parmon VN. Fast adsorptive and photocatalytic purification of air from acetone and dimethyl methylphosphonate by TiO₂ aerosol. *Appl Catal B-Environ* 2009; 89: 602-12.
- [67] Hager S, Bauer R. Heterogeneous photocatalytic oxidation of organics for air purification by near UV irradiated titanium dioxide. *Chemosphere* 1999; 38: 1549-59.
- [68] Hofstadler K, Bauer R, Novalic S, Heisler G. New reactor design for photocatalytic wastewater treatment with TiO₂ immobilized on fused-silica glass fibers: photomineralization of 4-chlorophenol. *Environ Sci Technol* 1994; 28: 670-4.